

Lecture #25

Date: Nov 25, 25

Day: Tuesday

Kruskal's Algorithm

- make each vertex (Minimum Spanning Tree)
- Sorts the edges ^{of its own set}
- For each edge:
 - If it connects 2 diff set → choose it
 - Union the set
- Continue until MST has $V-1$ edges.

$Kruskal(G, w)$

$A = \{ \}$

for each vertex " V " $\in G.V$
make-set (V)

// create a single list & store all the edges ... sort them in increasing order of weights of edges ... // sorting: $E \log E$
 $V^2 \log V^2$

for each edge (u, v) {

if ($find(u) \neq find(v)$) {

$A.insert((u, v))$; ^{if not from the same set (disjoint)}

Union (u, v)

}

}

return A;

}

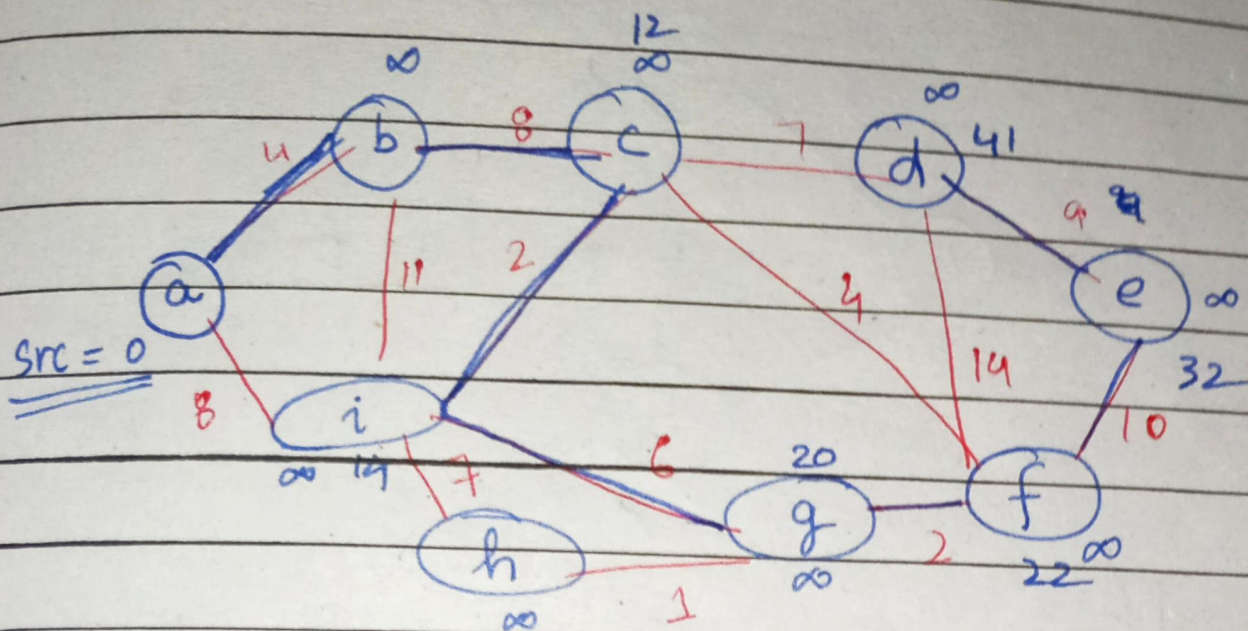
merging them into same set

$$T(N) = 2 V^2 \log V$$

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Prim's Technique



SPT
MST

Similar to Dijkstra:

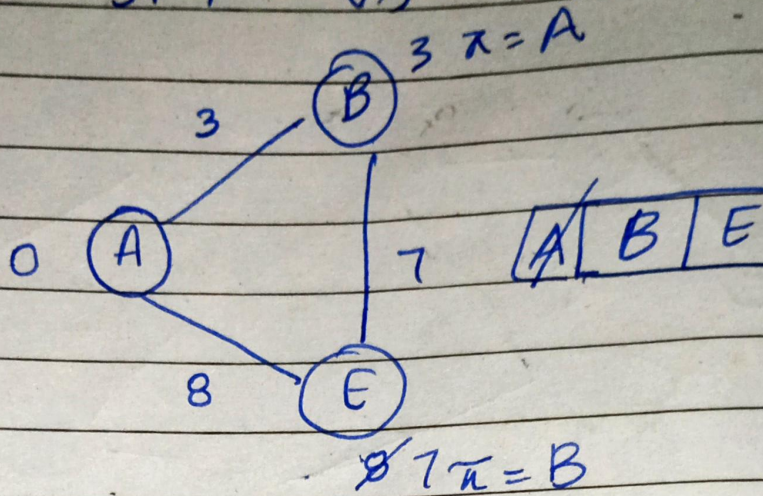
change in cost computation

if ($u \in Q$ && $u.cost > w(v, u)$) {
 $u.cost = w(v, u)$

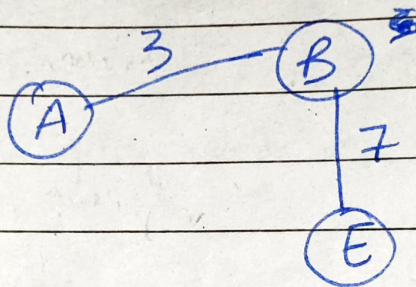
$u.\pi = v$
 }

SPT VS MST

connecting the whole network with min cost.

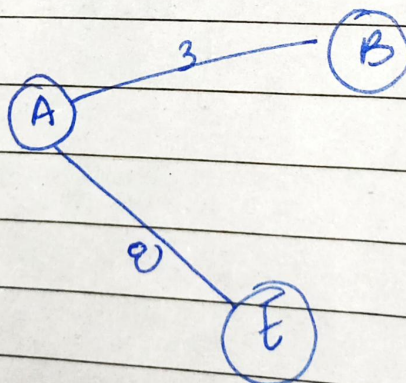


MST



∴ cost = 10

SPT:



∴ cost = 11

Lecture #26

Date: Nov 27, 25

Day: Thursday

All Pairs Shortest Path:

selecting intermediate nodes.

Floyd warshall Technique

for $k=1$ to $|G \cdot V|$ picking source

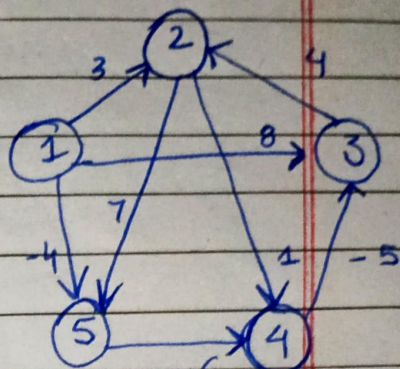
//src: for $u=1$ to $|G \cdot V|$

{ for $v=1$ to $|G \cdot V|$ }
dest. (all vertices)

$D[u,v] = \min(D[u,v], D[u,k] + D[k,v])$

}

intermediate vertex.



$D = w$ adjacency matrix.

for $k=1$ to $|G \cdot V|$

for $u=1$ to $|G \cdot V|$

for $v=1$ to $|G \cdot V|$

// intermediate

// source

// dest.

$D[u,v] = \min(D[u,v], D[u,k] + D[k,v])$

$$T(N) = O(N^3)$$

Adj	1	2	3	4	5
1	0	3	8	∞	-4
2	∞	0	∞	1	7
3	∞	4	0	∞	∞
4	2	∞	-5	0	∞
5	∞	∞	∞	6	0

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intermediate ver = 1

skip row
not
4 col
skip
diag

	1	2	3	4	5
1	0	3	8	∞	4
2	∞	0	∞	1	7
3	∞	4	0	∞	∞
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

$$\begin{aligned} D[4,5] &= \min(D[4,5], D[4,1] + D[1,5]) \\ &= \min(\infty, 2 + (-4)) \\ &= -2 \end{aligned}$$

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	1	2	3	4	5
1	0	3	8	4	4
2	∞	0	∞	1	7
3	∞	4	0	5	11
4	2	5	-5	0	-2
5	∞	∞	∞	6	0

$$\begin{aligned} D[1,4] &= \min(D[1,4], D[1,2] + D[2,4]) \\ &= \min(\infty, 3 + 1) = 4 \end{aligned}$$

$$\begin{aligned} D[3,5] &= \min(D[3,5], D[3,2] + D[2,5]) \\ &= (\infty, 4 + 7) = 11 \end{aligned}$$

$$\begin{aligned} D[3,4] &= \min(D[3,4], D[3,2] + D[2,4]) \\ &= \infty, 4 + 1 = 5 \end{aligned}$$

Shortest distance b/w 'u' to 'v'
must pass through 'w'
2-time Dijkstra Apply -

$$\begin{array}{ll} u \rightarrow w & : \sqrt{2} \log V \\ w \rightarrow v & : \sqrt{2} \log V \end{array}$$

$$\min = w\text{-cost} + v\text{-cost}$$

DFS - Application -

Bridge: edge that
joins two components..

Articulation points:

pt that connects the graph...
& by removing, components increase.

Approach 1 :- Remove one by one vertex
from the graph then apply DFS...
check component no.'s.

Approach 2: (Bridge) :-

delete edges one by one, check
components; if more than 1... then
it is a bridge.